Image captioning using CNN and Transformers

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***Abstract*—This research paper presents a method for image captioning where we combine models such as Convolutional Neural Networks (CNN) for visual feature vector extraction with Transformers for text sequence generation. The model aims to generate accurate and contextually meaningful captions for images. We employ the Flickr8k dataset for training and evaluation. The proposed architecture uses a pre-trained VGG16 CNN to extract image embeddings, while the transformer-based decoder generates captions by attending to relevant visual features. Experimental results demonstrate the model's effectiveness in generating fluent and coherent captions, achieving promising results on standard evaluation metrics.**

***Index Terms*—Image Captioning, Convolutional Neural Networks (CNN), Transformers, Multi-Head Attention, Positional Encoding, Sequence Generation, VGG16 Model, Computer Vision, Natural Language Processing (NLP), Flickr8k Dataset, Deep Learning, Image Feature Extraction, Caption Generation, Categorical Cross-Entropy, BLEU Score, CIDEr Score, ROUGE Score, METEOR Score, Parallelized Processing, Visual-Semantic Alignment**

I. Introduction

Image captioning is a critical and important task in computer vision (CV) and natural language processing (NLP) that involves generating descriptive captions from visual content. It has numerous applications, including automatic image tagging, assistance for visually impaired individuals, and image-based content retrieval.

Traditional image captioning models relied heavily on Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks. However, these models faced limitations in handling long-term dependencies and parallel processing. Recent approaches have adopted Transformer architectures to address these challenges, which utilize multi-head self-attention mechanisms for efficient and accurate caption generation.

In this paper, we propose a hybrid architecture combining VGG16 CNN for feature extraction and Transformers for sequential text generation. The model is trained on the Flickr8k dataset, which contains 8,000 images, each each image is paired with five descriptive captions of text of medium length.

II. Literature Review

Early image captioning models primarily used template-based methods or shallow learning techniques. With the advent of deep learning, CNN-RNN hybrid models became the standard, as they combined powerful visual feature extraction with sequential text generation. However, RNNs suffered from issues like vanishing gradients and slow training times.

The introduction of Transformers, that were introduced by Vaswani and others in 2017, revolutionized and made a significant impact in the field of sequence generation. Instead of using methods like recurrence which were traditional, self-attention mechanisms were used instead. This allowed faster processing of tasks and produced better results. Further, Vision Transformers (ViT) that were proposed by Dosovitskiy and others in 2020, used these attention-based techniques for visual tasks and achieved great results in the field. The project [1] addresses the challenge of accurately labeling satellite imagery affected by atmospheric conditions and diverse land cover types. To improve global understanding of deforestation—specifically where, how, and why it occurs—the researchers developed an advanced algorithm leveraging deep learning.

Using high-resolution satellite data, the system employs deep convolutional neural networks (CNNs) trained to detect multi-class, multi-label land cover changes. The model com- bines a pre-trained VGG-19 encoder (for feature extraction) with a GRU decoder and techniques like sparse cross-entropy loss to enhance prediction accuracy.

With deforestation claiming nearly 20% of the Amazon rainforest over four decades, this tool aims to provide precise, scalable monitoring of forest loss, supporting better conservation efforts worldwide.

In this work, we adopt a CNN-Transformer hybrid approach, leveraging CNNs for visual feature extraction and Transformers for efficient and contextually rich caption generation.

Image captioning is a deep learning task that combines computer vision (to understand images) and natural language processing (to generate text descriptions for the images). Although progress has been made, achieving human-like accuracy remains a challenge.

This study [2] systematically reviews deep learning-based methods for image captioning, analyzing 61 key studies (2017-2019) from databases such as Scopus, Web of Science and IEEEXplore. Key findings include:

- Model Architectures:
  - CNNs (e.g., ResNet, VGG) extract visual features.
  - RNNs/LSTMs generate captions, with LSTMs outperforming RNNs.
- Data sets:

  - MS COCO (most widely used), followed by Flickr8k and Flickr30k.
- Evaluation Metrics:
  - BLEU (1-4) is the standard for assessing caption quality.
- Best-Performing Methods:
  - Encoder-Decoder frameworks paired with attention mechanisms yield the most promising results.
  - The research serves as a guide for future work, suggesting that combining CNNs, LSTMs, and attention mechanisms can significantly improve auto-captioning accuracy.

Recent advances in artificial intelligence have placed image captioning as a important research area at the intersection of computer vision and natural language processing. This technology automatically generates natural language descriptions of visual content, representing a significant step toward comprehensive scene understanding.

The study [3] examines current methodologies with particular emphasis on attention mechanisms, which have become fundamental to modern image captioning systems. These mechanisms demonstrate remarkable effectiveness in bridging visual perception and linguistic expression. Key aspects covered in the review include:

- Detailed analysis of already existing techniques and their performance
- A thoughtful evaluation of both the strengths and weaknesses of already existing approaches
- The common benchmarks and evaluation metrics used in the field
- Finding useful datasets for training and validation of the models

The field of image captioning had always been using Recurrent Neural Networks (RNNs) in the past to generate descriptive text from images. Although, this study uses a Convolutional Neural Network (CNN) which is a relatively fresh method. It is specifically designed for language tasks. Unlike the RNNs, this can handle words one at a time and has lesser understanding of the larger context. The new Language CNN instead processes entire sequences of all the generated words at once. And as a result, it can understand the overall context better. This is essential for creating precise and accurate captions. By focusing on complete words sequences instead of the individual places, the model can successfully overcome the shortcomings of dealing with long-range dependencies, that are well known for troubling traditional methods.

Detailed evaluations on common benchmarks like Flickr30k and MS COCO show that the CNN-based architecture out performs the standard RNN models and provides results at par with the advanced modern state-of-the-art systems. Thus, it is demonstrated by the results that CNNs are highly effective for the sequence modelling tasks. They also make way for new opportunities for enhancing the multimodal AI applications. This discovery also points to promising prospects in the field of research for improving language generation in Computer Vision. But it also ensures that captions remain grammatically correct and contextually meaningful. Overall, this unique approach shows how neural networks can handle the limitations of traditional methods, and can lead to more dependable and adaptable solutions for various image captioning tasks.

Image captioning is a really challenging task, which involves creating precise and meaningful text descriptions of images. It requires identifying objects, their actions, their interactions, their complex relationships, and even the microscopic contextual details. This process combines Computer Vision methods to analyse visual content with natural language processing techniques, so that we can generate clear and grammatically correct captions. While machines are still far away from achieving human level understanding, deep learning methods have made significant improvements. Es- pecially, Convolutional neural networks (CNNs) are used to extract the visual features. On the other hand, Long-Short Term Memory (LSTMs) networks are used for generating text. These advancements have made it possible to use image captioning in practical applications like smart systems and IoT devices.

This survey paper [5] deeply looks at various methods used in image captioning such as retrieval-based, template-based and deep learning approaches. This also reviews the evaluation metrics commonly used to measure their performance. This paper provides valuable insights into the current advancements in this field and also highlights the future prospects for research.

Another paper [6] focuses on the deep learning approaches for automated image caption generation by introducing a custom ensemble model. This model combines an Inception-based CNN for extracting the visual features, along with a two-layer LSTM network for generating language. This whole architecture works by having the CNN encoder for the image processing tasks. While the LSTM decoder analyzes the patterns amongst the captions. The outputs are then merged together using dense layers to produce the final caption output. For comparison, various other architectures were also tested. This included those using GRU and Bi-directional LSTM networks. These models were trained on the Flickr8k data set where the text components were processed with GloVe word embeddings, and meaningful word representations were created. While testing, new captions were generated by analysing input images through the trained network. BLEU metrics were used for evaluating the performance and it showed great results. The Ensemble model achieved a BLEU-4 score of 55.8%. The GRU and Bi-directional LSTM implementations also produced different levels of effectiveness in the image captioning tasks.

This study [7] shows us an image captioning system that is highly advanced. It has the ability to perform image processing tasks and it generates three different outputs. These include output formats like text descriptions in multiple languages, audio files in MP3 format and different visual outputs. The system's implementation merges the capabilities of the Convolutional Neural Networks (CNNs) for visual features extraction

and long short-term memory (LSTMs) networks for language generation. This unique approach is successful in combining and connecting the fields of computer vision and natural language processing. The CNN encoder reviews the input images by comparing them with a large training dataset, so that it can identify and extract important visual details. While, the LSTM decoder uses these extracted features to create clear and logically coherent text descriptions. This novel design allows an easy integration of visual content analysis and language generation.

For comprehensive performance evaluation, the system utilizes the BLEU metric algorithm to quantitatively assess the quality and accuracy of generated captions against established benchmarks. This integrated approach demonstrates significant capabilities in producing accurate, multimodal descriptions through sophisticated deep learning techniques that simultaneously handle visual analysis and language generation tasks. The model's ability to deliver outputs in multiple formats makes it particularly suitable for diverse applications requiring both visual and linguistic interpretation of image content.

The paper [8] presents a novel Image Transformer architecture designed to address the challenges of applying transformer models to image captioning tasks. While transformers have proven highly effective for text-based applications, their direct use for visual captioning remains problematic due to fundamental differences in how linguistic elements (discrete words) and visual components (image regions) are structured. The proposed solution introduces two key innovations: a modified text encoding transformer and an text decoding transformer that incorporates the relative spatial relationships between vari- ous image regions. This architectural redesign features a wider layer structure specifically optimized for processing visual data while preserving the transformer's powerful attention mecha- nisms. The model demonstrates exceptional performance using only region features as input, achieving excellent results on both MSCOCO offline and online benchmarking tests. These findings validate the effectiveness of the architectural adaptations in filling the gap between visual understanding of images and language generation, while maintaining the transformer's core capabilities for attention-based processing. The success of this approach suggests new directions for adapting transformer architectures to multimodal tasks that require combining visual and linguistic information processing.

This paper[9] introduces CPTR (CaPtion TransformeR), a completely novel fully transformer-based architecture for image to text conversion that fundamentally rethinks the conventional "CNN+Transformer" paradigm. The proposed approach processes sequentialized raw image patches directly through the transformer encoder, enabling global context modeling from the initial encoding stage while completely eliminating convolutional operations. This pure transformer design demonstrates superior performance over existing hybrid CNN-Transformer methods on the MSCOCO benchmark dataset. The architecture's end-to-end attention mechanism provides interpretability advantages, allowing clear visualization of both intra-image patch relationships in the encoder and cross-modal word-to-patch attention patterns in the decoder. These visualizations provide deep insights into how the model's caption generation works. This also shows us how logical connections between visual details and language elements are made. The success of CPTR suggests that a unified transformer framework can effectively replace the traditional two-stage visual processing and language generation pipeline, while providing both performance improvements and greater architectural transparency.

This paper [10] presents a comprehensive empirical study of hyperparameter optimization in visual attention architectures for image captioning, systematically evaluating various loss functions and optimization methods to determine their im- pact on model performance. The research demonstrates that selecting cross-entropy as the main loss function combined with the Adam optimizer yields optimal results, achieving a top accuracy of 73.092% and the BLEU-4 score of 20.10. The study further compares the different convolutional neural network architectures as encoders, identifying ResNext-101 as the top performer with a top accuracy of 73.128% and BLEU-4 score of 19.80, while recognizing MobileNetV3 as a computationally efficient alternative that maintains reasonable accuracy with significantly lower resource requirements. The investigation extends to transformer-based architectures, revealing that the Data-efficient Image Transformer (DeiT) outperforms the Vision Transformer (ViT) with a substantially higher BLEU- 4 score of 34.44, suggesting promising directions for future architectural development in image captioning systems. These findings offer useful insights into balancing both performance and computational efficiency when applying visual attention mechanisms for generating captions.

This paper [11] provides a detailed overview of transformer-based methods for image captioning, which is a crucial task that bridges computer vision and natural language process- ing. The study explores how transformer architectures have evolved, starting from their initial use in natural language processing and successfully adapting them for tasks that combine vision and language. The authors carefully analyze various transformer-based captioning models, with an empha- sis on recent advancements using vision-language pre-training techniques that have delivered cutting-edge performance.

This review digs deep into the intricacies of the transformers architecture. It is focused on the attention mechanisms and how they have been adjusted accordingly for the visual comprehension tasks. The authors have compared various different transformer-based models for image captioning and have also debated about their training methods, design choices and performance characteristics. Certain parameters, such as widely used evaluation metrics and popular dataset standards, are employed to assess the quality of generated image captions.

Through their review, the authors pinpoint significant challenges in existing methods, including difficulties in capturing detailed visual relationships and creating contextually rich descriptions. The paper lastly suggests future prospects for research. These include various research opportunities like advancements in multimodal pre-training strategies, developing

more efficient attention mechanisms, and improving the techniques for combining visual and language data. This review works as both a technical overview and a theoretical piece for researchers that look towards improving image caption generation systems.

This study [12] explores how Transformer-based image captioning systems can be better understood and evaluated, especially since there is the increasing complexity of these vision-language models. It focuses on creating tools to analyze how these models generate text descriptions from visual inputs. And, it particularly emphasizes on their attention mechanisms and how the image regions and the words they produce over time align together.

The researchers use attribution methods to visually show which parts of an image the model focuses on during the caption generation process. These visualizations help in providing an explanation about how different components of this system can handle visual information and produce specific words in the captions. Also, the paper introduces new metrics to measure how well the model produces its predictions with the respective image regions corresponding to them. It also offers some objective aspects to evaluate its accuracy and identify the issues like generating irrelevant or incorrect information (hallucination).

Experimental evaluation performed an examination and show a contrast and comparison among the three different Transformer-based architectures. These include the traditional methods as well as the Vision Transformer (ViT) approaches for feature extraction. This analysis provides useful and valuable insights about how varying visual encoding techniques impact the model's ability to interpret and connect visual inputs with textual outputs. The proposed framework equips practitioners with practical tools to better understand model behavior, identify areas where models might fail, and enhance the reliability of image captioning systems. These improvements are especially important for real-world applications, where precise connections between visual elements and generated captions are essential.

This study [13] explores how Transformer-based image captioning systems can be interpreted, understood and evaluated. It also looks deeply into the increasing complexity of these models that combine vision and language tasks. The research is mainly focused on creating tools to analyze how these systems generate text descriptions from visual inputs. This primarily glances at their attention mechanisms and how image regions and words they produce align with each other over the period of time.

The authors use attribution methods to show which parts of an image the model focuses on during the caption creation process. These visualizations reveal how different architectural components process visual information when producing specific words in the output sequence. Additionally, the paper introduces novel metrics for quantitatively assessing the temporal alignment between model predictions and corresponding image regions, providing objective measures of a model's grounding accuracy and its tendency toward hallucination.

Experimental evaluations compare three distinct Transformer-based architectures, including both conventional and Vision Transformer (ViT) feature extraction approaches. This comparative analysis yields insights into how different visual encoding strategies affect the model's interpretability and grounding performance. The proposed framework offers practitioners valuable tools for diagnosing model behaviors, identifying failure modes, and ultimately improving the reliability of image captioning systems for real-world applications where accurate visual grounding is crucial.

III. Methodology

A. About the Dataset

We have used the Flicker 8K dataset. The Flicker 8K dataset is used as a benchmark in computer vision and natural language processing tasks. It contains around 8000 images that are taken from the Flicker website. Each of the images is paired with 5 captions that are written by human annotators. These captions provide a detailed description of images which eventually helps to train the models that include computer vision and natural language processing task. Each image in the dataset contains real world scenarios including people, place and things. The diversity in the dataset helps models to generalize over various situations. The availability of multiple captions for the single image enhances the model's ability to learn and generalize.

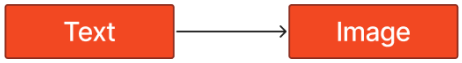

Fig. 1. Text to Image.

Flicker dataset is usually used to train the models that include both computer vision and natural language processing. The dataset can be used to generate the text using images (Image Captioning) and can be used to generate images from the text (Image generation from the textual prompt).

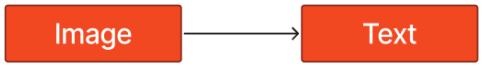

Fig. 2. Image to Text.

B. Data Preprocessing

We have used the flicker 8k dataset that contains 8000 images and each image has 5 captions for better understanding and generalization. We have used a pre-trained VGG-16 for getting the features from the images. Use of VGG-16 reduces the overall model size and training complexity by extracting the feature vectors beforehand. We have used VGG-16 to extract the feature of images. All the feature vectors are stored along with the image id in key value pairs after extraction form VGG-16. Features are stored in key value pair along

with image id to uniquely identify them while preparing the dataset for model training.

$$\text{Features} = \{\text{Image_ID}_i : \mathbf{F}_i \mid i \in \{1, 2, 3, \ldots, n\}\}$$

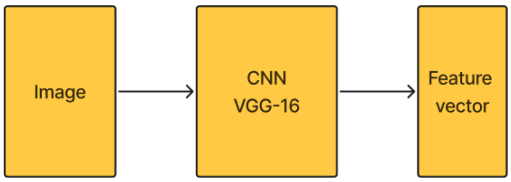

Fig. 3. VGG-16 for Feature Extraction.

The image feature vectors are stored as the key value pair in the dictionary data structure. Now all the captions are stored in the same key value pairs along with the image id.

$$\text{Captions} = \{\text{Image_ID}_i : \begin{bmatrix} \text{Caption}_1 \\ \text{Caption}_2 \\ \text{Caption}_3 \\ \text{Caption}_4 \\ \text{Caption}_5 \end{bmatrix} \mid i \in \{1, 2, 3, \ldots, n\}\}$$

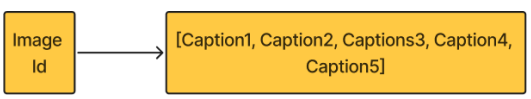

Fig. 4. Image Id to Caption.

All the captions are gathered in one place from the collected dataset. Each image has multiple annotated captions in the Captions data structure. Every key is an image ID and every value is a list of all the five captions. We have extracted all the captions in a single list and performed tokenization on the list. We have used the Keras Tokenizer to build the vocabulary. The tokenizer fits an entire collection of captions which ultimately converts all tokens into unique numerical values that are suitable for various deep-learning models. The size of the vocabulary can be calculated by the number of unique tokens after the tokenization step. To ensure the uniform length of all the text sequences. We have implemented the data generator functions that will acquire the feature vector of each Image and align it with corresponding input sequences of text that we have acquired after the tokenization step. Each input sequence is progressively built where the input sequence contains the tokens up to (i-) the index and the output sequence contains the word at ith index. All the input sequences are padded up to the maximum length of the sequences to ensure the uniform length of the input sequences. The image features (X1), partial input sequences (X2), and output word vectors (y) are appended until the batch size is reached. For every pair of partial input sequences and output word vector, we have its corresponding image feature vector.

$$\text{Partial Input Sequences} = \text{tokens}[: i]$$

$$\text{Output Word Vector} = \text{tokens}[i]$$

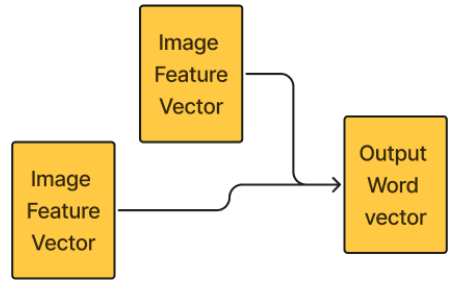

Fig. 5. VGG-16 for Feature Extraction.

C. Model Architecture

1. CNN-based VGG-16

The model helps us to get the feature vectors of the images. VGG is known for its depth, simplicity, and remarkable ability to extract all the important features from the image. Its ability to extract all the meaningful features from the image makes it the best choice. VGG 16 consists of 16 layers that include 13 convolutions and 3 fully connected layers at the end of the model layers. It uses a 3x3 convolutional filter with a stride of 1 and max pooling layers of 2x2 to reduce the spatial dimension. All the convolutional layers use ReLU as the activation function to introduce the non-linearity in the model. In our approach we are not classifying the images using VGG-16, we are using it for the deep feature extraction. We have used all of its convolutional layers excluding the fully connected layers that results in the feature extraction from the convolution layers. Before feeding images to the model, we have pre-processed them to match the input requirements of the model which is 224x224 pixels. After feeding the images to the VGG-16 model we have extracted the feature vector of size (1,4096) for each and every image.This feature vector will be used to fuse with the text input sequences so that the attention mechanism will be able to learn the representations from both the text and images.

Total params: 134,260,544 (512.16 MB)
Trainable params: 134,260,544 (512.16 MB)

2. Attention Mechanism

Attention mechanism is the fundamental concept in machine learning, especially in sequence modeling tasks such as NLP (Natural Language Processing) and CV (Computer Vision). The attention mechanism is inspired by human cognition where we selectively focus on the important parts of the sentence while ignoring the less important parts of the text. This approach helps models to learn faster and more ef- ficiently. Older Seq2Seq models often fail to maintain the

TABLE I
MODEL ARCHITECTURE DETAILS

Layer (Type)	Output Shape
Input Layer	(None, 224, 224, 3)
Block1 Conv1	(None, 224, 224, 64)
Block1 Conv2	(None, 224, 224, 64)
Block1 Pool	(None, 112, 112, 64)
Block2 Conv1	(None, 112, 112, 128)
Block2 Conv2	(None, 112, 112, 128)
Block2 Pool	(None, 56, 56, 128)
Block3 Conv1	(None, 56, 56, 256)
Block3 Conv2	(None, 56, 56, 256)
Block3 Conv3	(None, 56, 56, 256)
Block3 Pool	(None, 28, 28, 256)
Block4 Conv1	(None, 28, 28, 512)
Block4 Conv2	(None, 28, 28, 512)
Block4 Conv3	(None, 28, 28, 512)
Block4 Pool	(None, 14, 14, 512)
Block5 Conv1	(None, 14, 14, 512)
Block5 Conv2	(None, 14, 14, 512)
Block5 Conv3	(None, 14, 14, 512)
Block5 Pool	(None, 7, 7, 512)
Flatten	(None, 25088)
FC1 (Dense)	(None, 4096)
FC2 (Dense)	(None, 4096)

TABLE II
MODEL ARCHITECTURE DETAILS

Layer (Type)	Output Shape
InputLayer (input layer 2)	(None, 35)
Embedding (embedding)	(None, 35, 256)
NotEqual (not equal)	(None, 35)
MultiHeadAttention (multi head attention)	(None, 35, 256)
Add (add)	(None, 35, 256)
LayerNormalization (layer normalization)	(None, 35, 256)
Dense (dense 1)	(None, 35, 512)
InputLayer (input layer 1)	(None, 4096)
Dense (dense 2)	(None, 35, 256)
Dropout (dropout)	(None, 4096)
Add (add 1)	(None, 35, 256)
Dense (dense)	(None, 256)
LayerNormalization (layer normalization)	(None, 35, 256)
Add (add 2)	(None, 35, 256)
MultiHeadAttention (multi head attention 1)	(None, 35, 256)
Add (add 3)	(None, 35, 256)
LayerNormalization (layer normalization 1)	(None, 35, 256)
Dense (dense 3)	(None, 35, 512)
Dense (dense 4)	(None, 35, 256)
Add (add 4)	(None, 35, 256)
LayerNormalization (layer normalization 2)	(None, 35, 256)
Add (add 5)	(None, 35, 256)
Dense (dense 5)	(None, 35, 256)
Lambda (lambda)	(None, 256)
Dense (dense 6)	(None, 8485)

context as they compress all the input information in the fixed-size context vector that acts as a bottleneck. This bottleneck limits the model to learn and retain detailed information. These limitations can be overcome by taking the weighted sum of all encoder hidden states. It assigns the different weights or attention scores to different hidden states of the encoder. Attention Mechanism has revolutionized the way we capture the context from various parts of the input sequences. It can capture the long-range dependencies in the input sequences while preserving the context of the text. The attention mechanism attends to each and every token in the input sequence to get the context of the sequence.

- Self-Attention: It calculates the attention scores within the same sequence, capturing the long-range dependencies.
- Multi-Head Attention: This variant allows models to calculate the attention scores for more than one context in the given text.

3. Model Architecture:

The proposed model is designed to combine the multi-modal data. It combines the features of both text and image using Multi-Head Attention mechanism. This section explains about the overall architecture of the given model.
Total params: 10,202,405 (38.92 MB)
Trainable params: 10,202,405 (38.92 MB)

- Self-Attention: It calculates the attention scores within the same sequence, capturing the long-range dependencies.
- Multi-Head Attention: This variant allows models to calculate the attention scores for more than one context in the given text.
- Image features (Input 1): These features are extracted from the pre-trained Convolutional Neural Network modal that is VGG-16. We have passed the image data through VGG-16 for feature extraction and extracted (1,406) size feature vectors for each image. The image feature inputs go through the dropout regularization to mitigate overfitting. The dropout rate is 0.5 which means 50% of the neurons will be randomly dropped during training. The regularization helps the model to generalize on the unseen data and predict better for new inputs. The next layer reduces the dimensionality to 256 with non-linearity being introduced by the ReLU activation function.
  - Dropout (0.5): Helps prevent overfitting by randomly setting half the input units to zero.
  - Dense Layer (256 units): Projects high-dimensional input to a lower-dimensional space suitable for attention mechanisms.
- Text Sequences (Input 2): These are input sequences of tokens after the tokenization step. These tokens correspond to a single word in a vocabulary. The max sequence

length is the length of the largest sequence. These input tokens are now passed through the embedding layer that converts the integer-encoded words into a dense vector of size 256. Masing is enabled to allow the model to ignore the padding in the input sequences.

- Embedding Layer: Learns vector representations for words.
- Masking: Ensures that padding tokens do not contribute to the attention scores.

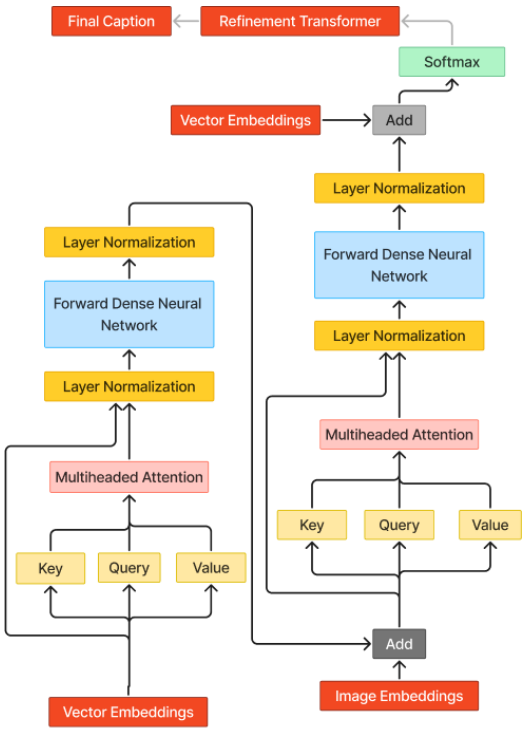

Fig. 6. Model Architecture.

4. Encoder

The Multi-Head Attention attend to different parts within the sequence simultaneously. In our implementation, we have used 8 attention heads each of dimension 256. These heads allow model the model to capture the multiple representation subspaces. This capability is essential for capturing the long-range dependencies and contextual relationships in the given data. By splitting attention into multiple heads, each head can focus on different contexts of the data enhancing the overall performance of the modal. These attention heads allow the model to grasp both local and global context.

1. Self Attention

Every token in the input sequence attends to every other word in the sequence to calculate the attention scores. This enables the model to capture the relationship between words called attention scores regardless of the distance between them.

2. Residual Connections

Here we add the input of the last layer with the output of the current layer helps the model to preserve the original information allowing for deeper architectures. The residual connections help to handle the vanishing gradient problem.

3. Lyer Normalization

Layer Normalization is applied to stabilize and speed up training. By using a very small epsilon value the model the model avoids division by zero. It is helpful in stable training by mitigating the risk of vanishing and exploding gradient problems. It does it by stabilizing the distribution of activations across layers resulting in smoother and more stable gradient flow.

4. Feedforward Network

The output from the attention layer is now being passed to the Normal Neural Network with two layers. The first layer is 512 units with a rectified linear unit activation function to introduce the non-linearity in the model. The next layer reduces the dimension of the previous layer to 256 units which results in the original dimensions that align with the input dimensions.

5. Cross Model Fusion

- Cross Model Fusion is a technique that is used to combine the information from various types of input data. Here we are combining the textual data and image data to capture the relationship between the image and the text.
- In our approach, we have used the element-wise addition to fuse the output of the encoder's attention layers and the image feature vectors. This helps us to add the complimentary information form both sources without introducing much complexity in the model.
- This operation of element-wise addition is simple yet effective since it does not use any complex and learnable parameters in this step. Its simplicity makes it computationally efficient.
- Cross-model fusion helps the model retain the information from both sources. Element-wise addition ensures that the model is not being dominated by one of the input features.

6. Decoder

The decoder block applies Multi-Head Attention Mechanism on the fused representation of decoders output and image vectors. This step is crucial as it refines the integrated feature by attending the most relevant features in the fused vector representations. The training the decoder on combines data helps the modal to learn from both images and text. We have used 8 attention heads, each with a key dimension of 256. The exact same architecture is being used in the encoder but here we are using it on the fused data.

7. The Final Integration

The final stage involves the integration of decoders output and original image vector. The step is essential for combining the input image features and decoders output before the final prediction is being made. A layer of 256 units is used to introduce the non-linearity. The lambda layer is introduced to get the final timestamp of the output.

1. Lambda Layer

It is used to extract the last timestep output from the sequence. It is used to extract the most relevant information that is present in the output layer.

2. Softmax Layer

It converts the final output into the probability over the vocabulary. The sum of all these probabilities is 1.

8. Model Compilation

This combines all the model components together into trainable architecture. Use of Adam optimizer ensures the adaptive learning rates and for efficient training. We have selected the Categorical cross entropy for the loss function due to its suitability for multiclass classification.

1. Adam Optimizer

Combines the benefit of RMS Prop and momentum that results in adaptive learning. It helps in handling sparse gradients and non-stationary objectives.

2. Categorical Cross Entropy

This loss function is ideal for multiclass classification and measuring the error between predicted class and actual class.

IV. Results and Discussion

In this work, we have combined the Convolutional Neural Network (CNN) with the Attention-Mechanism-based Transformer model. The model leverages the pre-trained VGG-16 for feature extraction and the transformer-based architecture model to convert these feature vectors to contextually relevant captions. The VGG-16 is a pre-trained image classification model known for its ability to learn complex patterns in the images. When the image is passed through the convolutional layers of VGG-16 the final dense vector representations of these images can be obtained.

For the task of generating captions, we have employed the transformer-based Encoder and Decoder architecture. For the given task of caption generation, we used a transformer-based architecture consisting of an Encoder and Decoder. This approach mainly utilizes a Multi-Head Attention mechanism to understand the complex long-range relationships in the text data. The encoder block incorporates an attention and residual connections to effectively learn from the input sequences. The image feature vector is then combined with the encoder's output and passed to the decoder. The decoder uses both the encoder's output and the image feature vector to predict the final sequence of text captions.

We evaluated the model's performance using the BLEU score (Bilingual Evaluation Understudy). The BLEU score is an evaluation metric which evaluates how closely the generated captions are similar with the original ones. This score helps measure the model's capability to create captions that are both grammatically correct and contextually meaningful. We had also performed a detailed overall examination to understand and interpret the uniqueness, clarity and logical meaningfulness of the generated captions in the specific context. This approach allowed us to identify the strengths and weaknesses of the model in practical use cases. Our findings demonstrate that the combination of CNN based feature extractor and a transformer-based caption generator is a powerful combination to produce the captions or descriptions from images. This type of model has multiple real-life applications. The model showcases the importance of Computer Vision and Natural Language Processing to address real-world problems.

1. BLEU Score

BLEU score is a technique in the natural language processing to compare the LLM generated text with the human generated text. Our model has achieved the following BLEU scores.

TABLE III
BLEU Score Evaluation

BLEU-1	BLEU-2	BLEU-3	BLEU-4
0.72	0.61	0.52	0.46

The gradual decline in the BLUE score across the higher-order score blue score is reasonable. A higher order BLUE score means the model is performing well for longer n-grams which are harder to match. The BLUE 4 reflects the balance between fluency and accuracy. The overall BLUE score shows that our model can produce meaningful text outputs for the given images.

2. Effectiveness of CNN (VGG-16)

The use of VGG-16 as the feature extractor played a significant role in extracting the feature vectors. With the use of VGG-16 as the feature, the extractor helped us to improve the overall model's accuracy. VGG-16 has provided the contextual rich dense vector for each image in the given dataset. During our evaluation, we observed that VGG-16 has captured fined-grained details such as

- Object Localization: Differentiating between foreground and background objects.
- Texture and Color: Highlighting features like the color of clothing or the texture of surfaces.
- Scene Context: Differentiating between indoors and outdoors

3. Impact of the Transformer Architecture

The Decoder part of the model plays a significant role in combining the information from both images and textual data. The use of a multi-head attention mechanism focuses on different parts of combined image and text sequences. Muti-head attention helps the model to learn from image and text data while at the same time effectively capturing the relationship between image features and text sequences.

4. Qualitative Analysis

To cross-check the correctness of the generated captions we have manually reviewed many of them. An example of a few captions is given.

Example1:
Image: The boy is playing with the dog in the park.
Ground Truth: A young boy is playing ball fetch with the brown dog on the ground.
Generated: A boy throws a ball towards his dog on the ground.

Example2:
Image: A group of people sitting at a dining table.
Ground Truth: Few people are having a dinner at the dinner table.
Generated: People are sitting around the table and having a meal.

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